

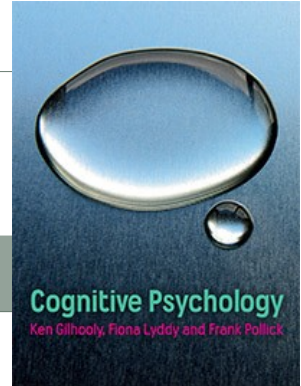
CHAPTER 4

LANGUAGE



CHAPTER 4

INTRODUCTION



LANGUAGE AND SOUNDS

Rule System	Description	Examples
Phonology	The <u>sound</u> system of a language. A <u>phoneme</u> is the smallest sound unit in a language.	The word <i>chat</i> has three phonemes or sounds: /ch/ /ă/ /t/. An example of phonological rule in the English language is while the phoneme /r/ can follow the phonemes /t/ or /d/ in an English consonant cluster (such as <i>track</i> or <i>drab</i>), the phoneme /l/ cannot follow these letters.
Morphology	The system of <u>meaningful</u> units involved in word formation.	The smallest sound units that have a meaning are called morphemes, or meaning units. The word <i>girl</i> is one morpheme, or meaning unit; it cannot be broken down any further and still have meaning. When the suffix <i>s</i> is added, the word becomes <i>girls</i> and has two morphemes because the <i>s</i> changed the meaning of the word, indicating that there is more than one girl.
Syntax	The <u>system</u> that involves the way words are <u>combined</u> to form acceptable phrases and sentences.	Word order is very important in determining meaning in the English language. For example, the sentence "Sebastian pushed the bike" has a different meaning than "The bike pushed Sebastian."
Semantics	The system that involves the <u>meaning</u> of words and sentences.	Knowing the meaning of individual words—that is, vocabulary. For example, semantics includes knowing the meaning of such words as <i>orange</i> , <i>transportation</i> , and <i>intelligent</i> .
Pragmatics	The <u>system</u> of using appropriate <u>conversation</u> and knowledge of how to <u>effectively</u> use language in <u>context</u> .	An example is using polite language in appropriate situations, such as being mannerly when talking with one's teacher. Taking turns in a conversation involves pragmatics.

- **Phones**
 - basic speech sounds (about 100)
 - Phonetics = study of the speech sounds.
- **Phonemes**
 - smallest meaningful sound unit.
 - Differences between languages! E.g., Finnish 56, Spanish 24
- **Allophones**
 - phonetic variants of the same phoneme.
- **Phonotactic rules**
 - the combination of sounds permitted in a language
 - E.g., in English /ng/ in end of word ok, not possible at beginning

LANGUAGE AND COMMUNICATION

- Communication
 - = any means by which information is shared
 - = a source encodes and transmits a signal, which is detected by a receiver and decoded in meaningful terms
- Language is more than communication (but is core feature)
- Two ways we can use language to communicate:
 - Written language
 - most recent 5000 years
 - allows a language to survive.
 - Speech
 - principal way we use language
 - existed for tens of thousands of years
 - qualitatively different from animal communication.
- + Non-verbal communication

LANGUAGE UNIVERSALS

- 6,000 languages in use.
- Vary in phonetics, syntax and lexicon.
- Linguistic universals or features found in all languages:
 - have consonants and vowels
 - combine basic sounds into larger units
 - have nouns (words for people, places and objects, e.g. book)
 - have verbs (words that represent actions or 'doing', e.g. to read)
 - can combine words in meaningful ways
 - can express who did what to whom
 - can express sentences as negatives
 - can express sentences as questions
 - are structure-dependent, that is, involve a syntactic structure or grammar
 - allow recursion (the use of a rule within itself, allowing, for example, embedded sentences).

LANGUAGE UNIVERSALS

- 6,000 languages in use.
- Vary in phonetics, syntax and lexicon.
- Linguistic universals or features found in all languages:
 - ~~have consonants and vowels~~ Sign languages
 - ~~combine basic sounds into larger units~~
 - have nouns (words for people, places and objects, e.g. book)
 - have verbs (words that represent actions or 'doing', e.g. to read)
 - can combine words in meaningful ways
 - can express who did what to whom
 - can express sentences as negatives
 - can express sentences as questions
 - are structure-dependent, that is, involve a syntactic structure or grammar
 - allow recursion (the use of a rule within itself, allowing, for example, embedded sentences).

LANGUAGE UNIVERSALS

- 6,000 languages in use.
- Vary in phonetics, syntax and lexicon.
- Linguistic universals or features found in all languages:
 - have consonants and vowels Some nouns are actions/ occurrences, like *destruction*
 - combine basic sounds into larger units
 - have nouns (~~words for people, places and objects, e.g. book~~)
 - have verbs (words that represent actions or 'doing', e.g. to read)
 - can combine words in meaningful ways
 - can express who did what to whom
 - can express sentences as negatives
 - can express sentences as questions
 - are structure-dependent, that is, involve a syntactic structure or grammar
 - allow recursion (the use of a rule within itself, allowing, for example, embedded sentences).

LANGUAGE UNIVERSALS

- 6,000 languages in use.
- Vary in phonetics, syntax and lexicon.
- Linguistic universals or features found in all languages:
 - have consonants and vowels
 - combine basic sounds into larger units
 - have nouns (words for people, places and objects, e.g. book)
 - have verbs (words that represent actions or 'doing', e.g. to read)
 - can combine words in meaningful ways
 - can express who did what to whom
 - can express sentences as negatives
 - can express sentences as questions
 - are structure-dependent, that is, involve a syntactic structure or grammar
 - allow recursion (the use of a rule within itself, allowing, for example, embedded sentences).

Some nouns can also describe actions, like to text

THE EIGHT PARTS OF SPEECH

Category	Description
Verb	A very large class of words among which are many denoting actions (<i>run, jump, swim</i>) or states or experiences (<i>feel, see, think</i>). Auxiliary verbs mark tense, aspect and modality (<i>has eaten, is eating, may eat</i>). Verbs that take a direct object (as in <i>The dog ate the bone</i>) are called transitive verbs. Others are intransitive (<i>The dog slept</i>).
Noun	An extremely large class of words including the words denoting people, places, things, or abstract ideas in the broadest sense. The nouns include not only <i>desk, book, football, house</i> , etc., but also <i>absence, thought, possibility, failure</i> , etc. Proper nouns begin with a capital letter and name particular people, places, or things: <i>Barack Obama, London, Disneyland</i> . The pronouns are a special subclass of the nouns, used for referring to things without naming them (often because they are named elsewhere in the context): <i>he, she, I, you, they, it</i> .
Determinative	The fairly small class of determinatives includes words usually used with nouns to specify definiteness or quantity: <i>the, a, some, all, every, many, much, several, this</i> , etc. Sometimes they occur without nouns (<i>Some may disagree; I like this</i>), or in other uses (<i>So much the better</i>).
Adjective	The adjectives are a large class of words that typically modify or qualify the meaning of nouns, as <i>brown</i> does in <i>the brown dog</i> or <i>Our dog is brown</i> : words like <i>happy, cloudy, intelligent, mysterious, cool</i> , and many others.
Adverb	The great majority of the adverbs are formed from adjectives by adding <i>ly</i> : <i>nice</i> is an adjective and <i>nicely</i> is an adverb. They very commonly modify words other than nouns, providing information about where, when, how, or to what extent things happen: <i>locally, recently, awkwardly, extremely</i> . Adverbs unrelated to adjectives include <i>soon, quite, too, always, seldom</i> .
Preposition	Prepositions often precede noun phrases to indicate spatial or temporal relationships, as in <i>the book is on the shelf</i> . The prepositions include <i>after, at, before, by, despite, during, except, for, in, into, of, on, over, since, through, to, under, with, without</i> . Some occur before clauses (<i>after we met</i>), and some occur without anything following (<i>Go right in</i>).
Subordinator	There is a very small class of words called subordinators that function to mark clauses embedded in other clauses: <i>that</i> as in <i>that nobody loves me</i> ; <i>whether</i> as in <i>whether anybody loves me</i> ; <i>for</i> as in <i>for someone to love me</i> .
Coordinator	There is a very small class of words called coordinators, used to link clauses or phrases together: <i>and, or, but, nor</i> , and a very few others. (The subordinators and coordinators are traditionally classed together and called 'conjunctions'.)
Interjection	An interjection is a word that interrupts a sentence with an immediate expression of emotion or sentiment: <i>yikes, oops, shush, ouch</i> , etc.

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.
- Vocal-auditory communication channel
- Broadcast transmission and directional reception
- Rapid fading
- Interchangeability
- Feedback
- Specialization
- Semanticity
- Discreteness
- Displacement
- Productivity
- Cultural transmission
- Duality (of patterning)
- Prevarication
- Reflexiveness
- Learnability

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.
- Vocal-auditory communication channel
- Broadcast transmission and directional reception
- Rapid fading
- Interchangeability
- Feedback
- Specialization
- Semanticity
- bees and figure dancing

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.
- Vocal-auditory communication channel bees and figure dancing
⇒ Out from source (mouth),
- Broadcast transmission and directional reception localized in space by receiver
- Rapid fading
- Interchangeability
- Feedback
- Specialization
- Semanticity

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.
- Vocal-auditory communication channel bees and figure dancing
⇒ Out from source (mouth),
- Broadcast transmission and directional reception localized in space by receiver
⇒ written language
- Rapid fading ⇔
- Interchangeability
- Feedback
- Specialization
- Semanticity

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.
- Vocal-auditory communication channel bees and figure dancing
⇒ Out from source (mouth),
- Broadcast transmission and directional reception localized in space by receiver
⇒ written language
- Rapid fading ⇒ Sender can be receiver
- Interchangeability
- Feedback
- Specialization
- Semanticity

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.
- Vocal-auditory communication channel bees and figure dancing
⇒ Out from source (mouth),
- Broadcast transmission and directional reception localized in space by receiver
⇒ written language
- Rapid fading ⇒ Sender can be receiver
- Interchangeability ⇒ Speaker can monitor message
- Feedback
- Specialization
- Semanticity

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.
- Vocal-auditory communication channel
 - ⇒ bees and figure dancing
 - ⇒ Out from source (mouth), localized in space by receiver
- Broadcast transmission and directional reception
 - ⇒ written language
- Rapid fading
 - ⇒ Sender can be receiver
- Interchangeability
 - ⇒ Speaker can monitor message
- Feedback
 - ⇒ More energy ≠ other meaning
- Specialization
- Semanticity

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.
- Vocal-auditory communication channel
 - ⇒ bees and figure dancing
 - ⇒ Out from source (mouth), localized in space by receiver
- Broadcast transmission and directional reception
 - ⇒ written language
- Rapid fading
 - ⇒ Sender can be receiver
- Interchangeability
 - ⇒ Speaker can monitor message
- Feedback
 - ⇒ More energy ≠ other meaning
- Specialization
 - ⇒ Sounds in speech refer to objects and entities in world; words have meaning
- Semanticity

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.

Arbitrary relation word/referent $\ni$

- Discreteness
- Displacement
- Productivity
- Cultural transmission
- Duality (of patterning)
- Prevarication
- Reflexiveness
- Learnability

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.

Arbitrary relation word/referent	⊃	• Discreteness
Composed of discrete units	⊃	• Displacement
		• Productivity
		• Cultural transmission
		• Duality (of patterning)
		• Prevarication
		• Reflexiveness
		• Learnability

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.

Arbitrary relation word/referent	⊃	• Discreteness
Composed of discrete units	⊃	• Displacement
Not necessarily current context	⊃	• Productivity
		• Cultural transmission
		• Duality (of patterning)
		• Prevarication
		• Reflexiveness
		• Learnability

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.

Arbitrary relation word/referent	⊃	• Discreteness
Composed of discrete units	⊃	• Displacement
Not necessarily current context	⊃	• Productivity
Novel sentences	⊃	• Cultural transmission
		• Duality (of patterning)
		• Prevarication
		• Reflexiveness
		• Learnability

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.

Arbitrary relation word/referent	⊃	• Discreteness
Composed of discrete units	⊃	• Displacement
Not necessarily current context	⊃	• Productivity
Novel sentences	⊃	• Cultural transmission
Learned through interaction	⊃	• Duality (of patterning)
		• Prevarication
		• Reflexiveness
		• Learnability

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.

Arbitrary relation word/referent	⊃	• Discreteness
Composed of discrete units	⊃	• Displacement
Not necessarily current context	⊃	• Productivity
Novel sentences	⊃	• Cultural transmission
Learned through interaction	⊃	• Duality (of patterning)
Combi meaningless & meaning	⊃	• Prevarication
		• Reflexiveness
		• Learnability

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.

Arbitrary relation word/referent	⊃	• Discreteness
Composed of discrete units	⊃	• Displacement
Not necessarily current context	⊃	• Productivity
Novel sentences	⊃	• Cultural transmission
Learned through interaction	⊃	• Duality (of patterning)
Combi meaningless ↔ meaning	⊃	• Prevarication
Deceive and lie is possible	⊃	• Reflexiveness
		• Learnability

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.

Arbitrary relation word/referent	⊃	• Discreteness
Composed of discrete units	⊃	• Displacement
Not necessarily current context	⊃	• Productivity
Novel sentences	⊃	• Cultural transmission
Learned through interaction	⊃	• Duality (of patterning)
Combi meaningless ↔ meaning	⊃	• Prevarication
Deceive and lie is possible	⊃	• Reflexiveness
Communicate about language	⊃	• Learnability

HOCKETT'S DESIGN FEATURES FOR HUMAN LANGUAGE

- Charles Hockett's (1960)
 - 16 design features for human language
 - Animal systems share some of the features, but only human system demonstrates the full set.

Arbitrary relation word/referent	⊃	• Discreteness
Composed of discrete units	⊃	• Displacement
Not necessarily current context	⊃	• Productivity
Novel sentences	⊃	• Cultural transmission
Learned through interaction	⊃	• Duality (of patterning)
Combi meaningless ↔ meaning	⊃	• Prevarication
Deceive and lie is possible	⊃	• Reflexiveness
Communicate about language	⊃	• Learnability
Can be learned by speaker of another language	⊃	

OUR ADVANTAGE: ABILITY TO TRANSMIT ... (HARARI, 2011)

- larger quantities of information about the world:
 - Not: "*careful, lion*" or "*careful, eagle*" (e.g., green monkeys)
 - But: "*I saw this morning a lion near the bend of the river. The lion was tracking a herd of bison. She was heading towards ...*" So, deliberating (in group) how to chase away lion and hunt bison.
- larger quantities of information about social relationships:
 - Hence, not 20-50 in group (e.g. chimpanzees)
 - But, groups up to 150 individuals
- Information about things that do not really exist (spirits, nations, money, rights, ...):
 - Hence coöperation between large numbers of strangers and rapid innovation of social behavior

2. LANGUAGE ACQUISITION



HOW DO WE ACQUIRE THESE LANGUAGES?

- Most fundamental question about language acquisition:
 - To which extent is it the result of innate mechanisms or predispositions?
- This question is not a surprise:
 - When children learn a language, they learn the phonological, morphological, syntactic, semantic and pragmatic rules:
 - No one teaches them these rules explicitly
- Hence,
 - children just pick them up?
 - children have innate knowledge/mechanisms?
 - Especially important, because there is similarity in language acquisition!!

DARWIN ON LANGUAGE (1871)

86

The Descent of Man.

PART I.

sounds and ideas; and this obviously depends on the high development of his mental powers.

As Horne Took, one of the founders of the noble science of philology, observes, language is an art, like brewing or baking; but writing would have been a better simile. It certainly is not a true instinct, for every language has to be learnt. It differs, however, widely from all ordinary arts, for man has an instinctive tendency to speak, as we see in the babble of our young children; whilst no child has an instinctive tendency to brew, bake, or write. Moreover, no philologist now supposes

LINGUISTICS AND THE UNIVERSAL FEATURES OF LANGUAGE

- Innate abilities often share certain characteristics:
 - Present in all normal individuals
 - Uniform and automatic acquisition (same stages, without requirement of instructions)
 - Sometimes critical period for successful acquisition
 - Functionally and anatomically autonomous or modular from other abilities
 - Heritable

WALK VS KNIT

- Present in all normal individuals
W
- Uniform and automatic acquisition (same stages, without requirement of instructions) W
- Sometimes critical period for successful acquisition
- Functionally and anatomically autonomous or modular from other abilities W
- Heritable W



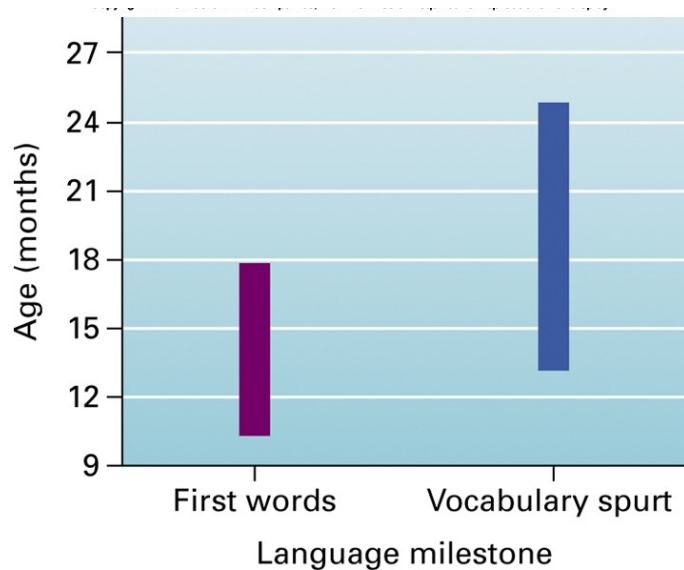
LINGUISTICS AND THE UNIVERSAL FEATURES OF LANGUAGE

- Learning to talk $\approx$ learning to walk
 - But, only one basic walk
- Children able to learn all human languages?
 - All human languages are remarkable similar to one another
 - Different theories different opinions about what is similar

UNIFORMITY IN LANGUAGE ACQUISITION

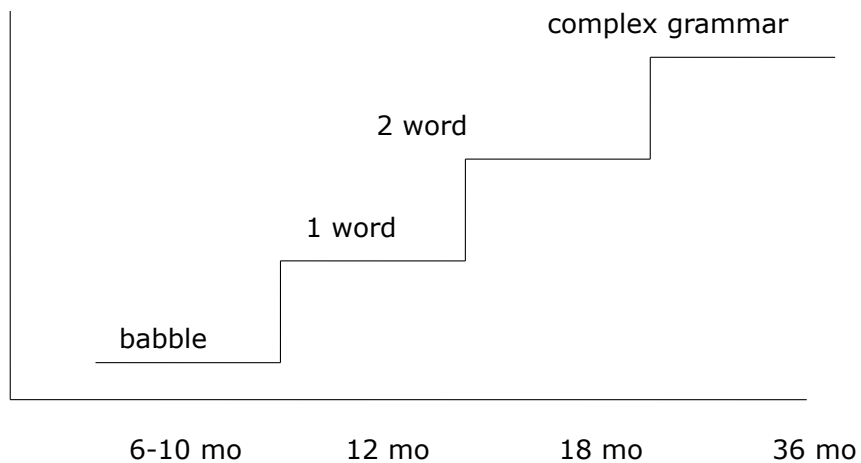
- Within a given language, the course of language acquisition is remarkably uniform:
 - First referential words at around 9-15 months
 - Next 6-8 months acquire single words in a fairly slow fashion until they have acquired approximately 50 words
 - After 50 words, rapid increase in vocabulary: 22-37 words per month (= new words on a daily basis)
- Lexical development in children is fast
 - 14,000 words by age 6

BEFORE THE FIRST WORDS: PRE-LINGUISTIC STAGE



- Birth - 6 weeks:
 - Crying
- 6 weeks - 4 months:
 - Cooing [vowel sounds only]
- 4 months - 7 months:
 - Babbling
 - much more variation
 - when baby is alone
 - Together with someone = protoconversation (turntaking!)

FROM BABBLE TO ONE-WORD



- Not always easy to determine
- Michael Frank, developmental psychologist:
 - Is also observing his daughter

MICHAEL FRANK, DEVELOPMENTAL PSYCHOLOGIST

- *M has been babbling ba, da, and ma for quite a while now.*
- *But at about 10.5 months, she started producing a very characteristic sequence: "BAba."*
- *This sequence showed up in the presence of "Brown Bear, Brown Bear, What Do You See," a board book by Eric Carle that she loves and that she reads often both at daycare and at home.*

IT IS SUBTLE (STAGER & WERKER, 1998)

- 14-month-olds:
 - could distinguish the sounds "bih" and "dih."
- However, they had trouble learning to pair these sounds consistently with different pictures:
 - However, they could do it with more dissimilar words like "lif" and "neem".

MICHAEL FRANK, DEVELOPMENTAL PSYCHOLOGIST

- Really a word? Yes, because:
 1. Consistency of form.
 - The intonation was descending, the stress was on the first syllable, and there was a hint of rounding ("B(r)Ab(r)a").
 - It felt very word-y.
 2. Consistency of context.
 - We heard this again and again when we would bring the book to her.
 3. Low frequency in other contexts.
 - We pretty much only heard it when "Brown Bear" was present, with the exception of one or two potential false alarms when another book was present.

BERGELSON & ASLIN (2017)

- Study with 6 months old babies, with eye-tracking and with video and audio recordings
- Babies as young as 6 months old may have an inkling/suspicion that certain words and concepts are related to each other.
 - Babies looked more at the image of the item that was mentioned by the parent when the other item on the screen was unrelated to it
 - If they look more at an image after it gets named than they did before they heard anything, they know [something about] what the word means
- Babies who were more often exposed to adults talking to them about items in their vicinity did better at identifying a picture of an object when the item was said out loud.

Table 8.2 Overview of Typical Language Development

Age	General Characteristics
Months	
1–5	<i>Reflexive communication:</i> Vocalizes randomly, coos, laughs, cries, engages in vocal play, discriminates language from nonlanguage sounds
6–18	<i>Babbling:</i> Verbalizes in response to speech of others; responses increasingly approximate human speech patterns
10–13	<i>First words:</i> Uses words; typically to refer to objects
12–18	<i>One-word sentence stage:</i> Vocabulary grows slowly; uses nouns primarily; overextensions begin
18–24	<i>Vocabulary spurt:</i> Fast-mapping facilitates rapid acquisition of new words
Years	
2	<i>Two-word sentence stage:</i> Uses telegraphic speech; uses more pronouns and verbs
2.5	<i>Three-word sentence stage:</i> Modifies speech to take listener into account; overregularizations begin
3	Uses complete simple active sentence structure; uses sentences to tell stories that are understood by others; uses plurals
3.5	<i>Expanded grammatical forms:</i> Expresses concepts with words; uses four-word sentences
4	Uses imaginary speech; uses five-word sentences
5	<i>Well-developed and complex syntax:</i> Uses more complex syntax; uses more complex forms to tell stories
6	Displays metalinguistic awareness

© 2007 Thomson Higher Education

RELATION BETWEEN DUTCH AND ENGLISH

The mean percentages words for each category

	American English	Dutch
Animals	7.4	9.6
Food/drink	10.0	6.7
Furniture	3.2	5.5
Vehicles	3.2	4.6
Toys	3.3	4.6
Clothes	3.4	3.0
Body parts	2.2	2.8

THE FIRST 50 WORDS IN DUTCH (SCHLICHTING, 1996)

Words, which were among the first 50 in at least 1/3 of the 37 children in the research ⇒ 36 words (from 389)

Nouns: car (35), mom (33), dad (32), cat (30), grandfather (27), cookie (23), grandmother (23), doll (22), ball (21), horse (21), dog (18), coat (17), clock (17), duck (16), apple (14), book (14), monkey (13), cheese (13)

Verbs: to eat (20), sit (17), sleep (15), poo (14), drink (13) **Adverbs:** out (21), on (20), open (13)

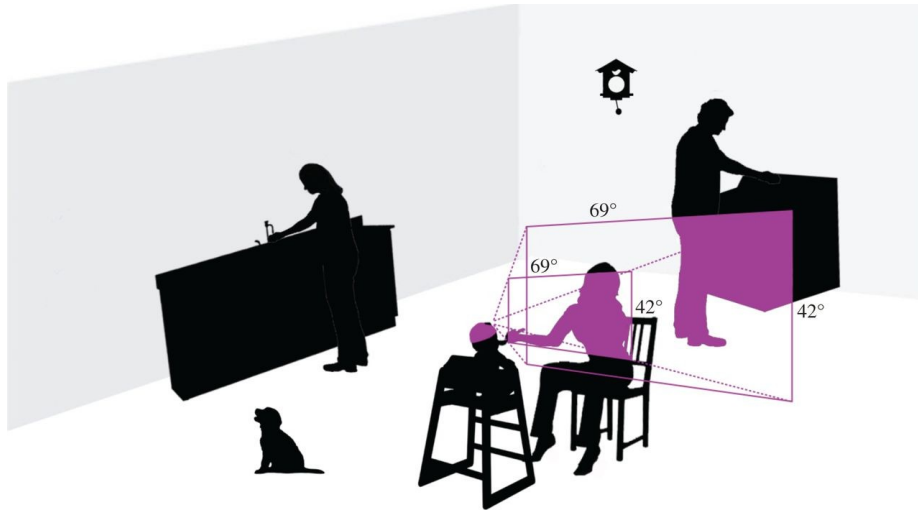
Social expressions: bah (29), hello (25), no (24), au (22), yes (21), boem (13), peep-bo (13)

Others: that (21)

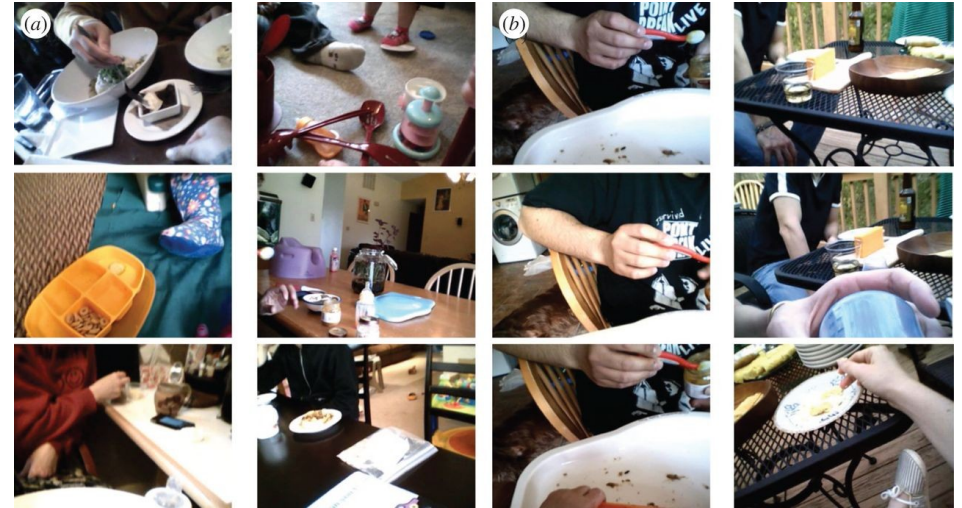
CLERKIN ET AL. (2016)

- Why these words?
- Images from head camera video captured by 8.5- to 10.5-month-old infants
 - 5 girls and 3 boys
 - Is the period before children engage in verbal interactions with parents and caregivers
 - cameras worn about 4.4 hours
- 147 at-home mealtime events were analyzed for the objects in view.

CLERKIN ET AL. (2016)



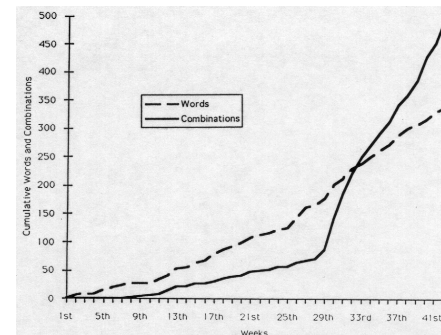
CLERKIN ET AL. (2016)



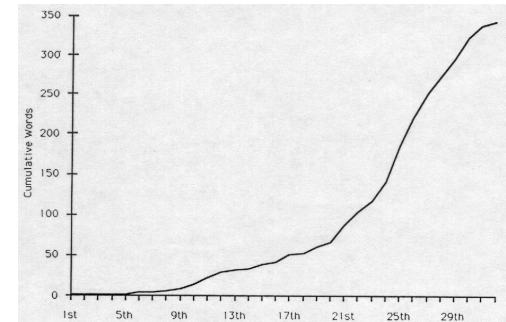
CLERKIN ET AL. (2016)

- Strong correlation between most frequently appearing objects and “first nouns” (= words acquired by half of all 16-month-olds; e.g., table, shirt, chair, spoon)
 - the number of times an object enters an infant’s field of vision helps associating certain words with certain objects
 - Hence, some of children’s first words are predictable based on their visual experience with objects and the prevalence of those objects in their visual world
- Hence, a developmentally informed conceptualization of statistical learning may emerge as the plausible mechanism through which infants learn their first object names.

NOT ALWAYS A VOCABULARY SPURT

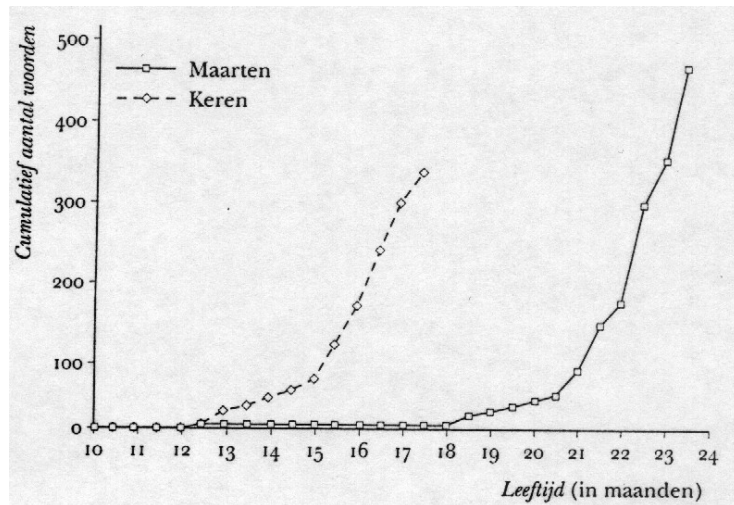


“Atypical child”: slow start, then no real spurt, a steady increase



“Normal child”: very slow start, slight increase, then strong increase

GIRL VS BOY



Girl earlier than boy in figure, but also a typical finding in literature!

- The vocabulary spurt begins when children recognize the symbolic nature of language
 - i.e. when they recognize that everything has a name.
- The vocabulary development is made possible by advanced articulatory skills.
- Big differences between kids and between production and comprehension:

MICHAEL FRANK, DEVELOPMENTAL PSYCHOLOGIST

- *Overall, I am still struck by how hard production is for M, compared with comprehension.*
- *A new word, say "playground" might start as something resembling "PAI-go" but merge back into "BA-ba" by the end of a few repetitions.*
- *M has never been a big babbler, and so I suspect that she is slow to produce language because the skills of production are simply not as well-practiced.*
- *There are some kids who babble up a storm, and I imagine all of the motor routines are much easier for them. In contrast, M just doesn't have the sounds of language in her mouth yet.*

MAYOR & PLUNKETT (2014)

- Data from 14607 infants and toddlers in five countries
- Investigated variability between individual lexicons during development for both comprehension and production.
- Early lexicons are highly overlapping.

MAYOR & PLUNKETT (2014)

- However, beyond 100 words:
 - toddlers share more words with other toddlers in comprehension than in production
 - even when matched for lexicon sizes!
- Hence, a structural difference in early comprehension and production:
 - Toddlers are generalists in comprehension
 - But they develop a unique, expressive voice.
- Variability in production decreases after 2 years:
 - convergence to a common expressive core vocabulary.

18 MONTHS AND LATER

- English is *morphologically impoverished* language.
- Words defy a simple definition:
 - English: a word is smaller than a sentence
 - Spanish: Hazmelo [do(haz) it(lo) for me(me)]
 - Swahili: atakusumbua [s/he will annoy you]
- Morpheme:
 - the smallest meaning unit within a word

SOME INFO ABOUT MORPHEMES

- Morphemes carry information about
 - Meaning (*bat*)
 - Function (*bat* + *s*)
- Meaning of individual morphemes should contribute to the overall meaning of the word: *pumpkin* ≠ *pump* + *kin*
 - Morpheme ≠ syllable!!

SOME EXAMPLES

- Mother = mother
- Eraser = erase + -er [*er* not always a morpheme]
- Jack's = Jack + -s
- Inabilities = able + in-, -ity, -s

INFLECTIONAL MORPHOLOGY

- Inflectional affixes mark tense, aspect, person, number,
- English has an impoverished inflectional system
 - It has only 8 inflectional suffixes

THE 8 ENGLISH INFLECTIONAL SUFFIXES

- | | | | |
|----|-------|-----------------|-----------------------------------|
| 1. | -s: | 3subject | <i>She wait-s for me.</i> |
| 2. | -ed: | past tense | <i>She wait-ed all day.</i> |
| 3. | -ing: | progressive | <i>She is wait-ing.</i> |
| 4. | -en: | past participle | <i>She has eat-en all of it.</i> |
| 5. | -s | plural | <i>She ate donut-s</i> |
| 6. | -s | possessive | <i>She ate John's donuts.</i> |
| 7. | -er | comparative | <i>She is short-er than John.</i> |
| 8. | -est | superlative | <i>She is the short-est girl.</i> |
- Other languages have a much richer system of inflection

18 MONTHS AND LATER IN ENGLISH (AND SIMILAR LANGUAGES)

- Infants form two-word utterances:
 - *Want cookie, give dolly, he big,...*
- Precursors of sentences/legitimate portions of sentences:
 - Not: *cookie want, big he*
- Two-word stage gradually becomes more complex: *telegraphic speech*:
 - Multiple word utterances
 - lack of inflectional or functional elements
 - *Anna want cookie, milk all gone, ...*
- By 3-4 years old, vast majority of sentences are grammatical

18 MONTHS AND LATER IN TURKISH (AND SIMILAR LANGUAGES)

- Use functional category morphemes at a younger age:
 - Often morphologically complex words before multi-word utterances
- Hence, for some aspects clear differences between languages